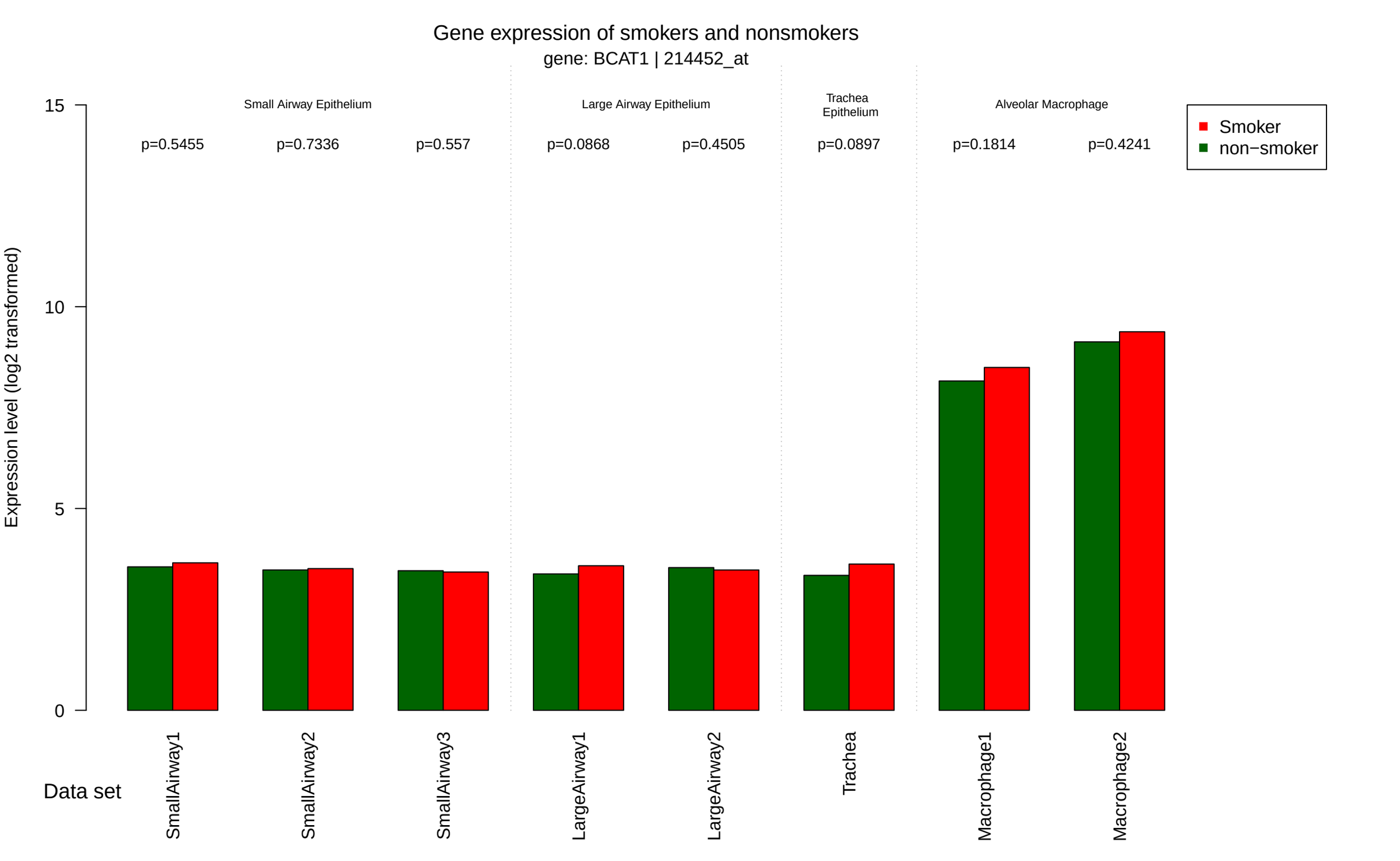
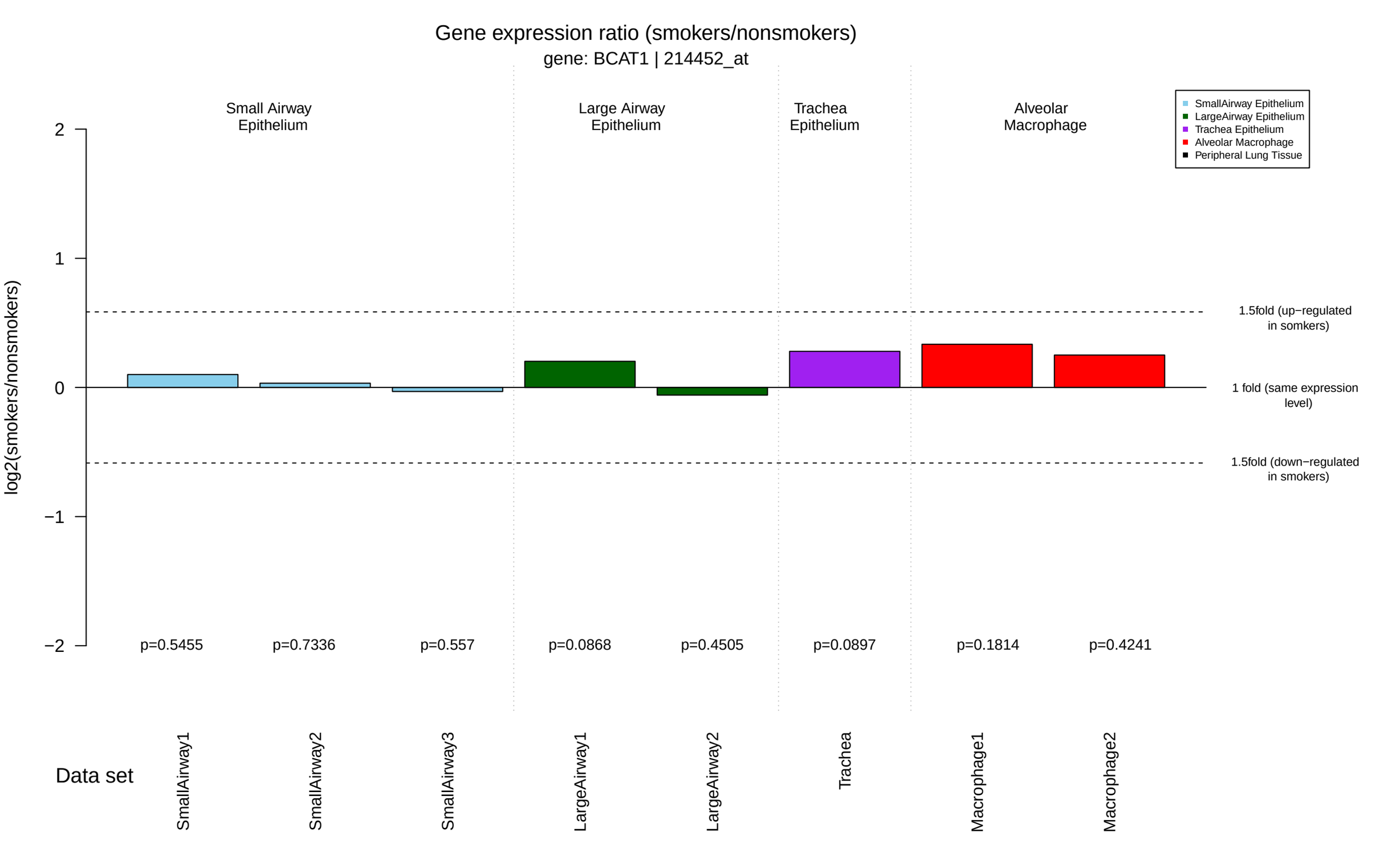
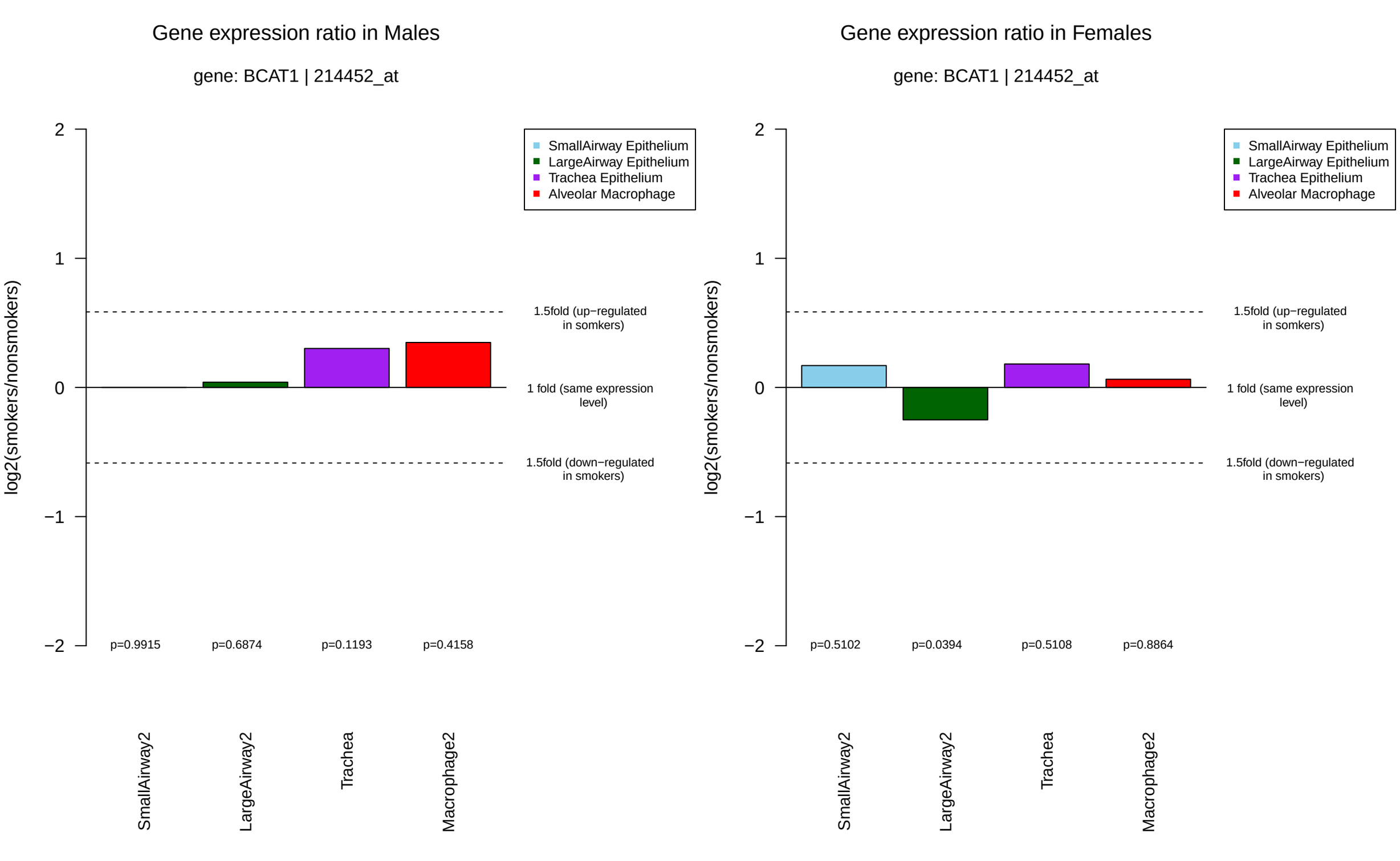
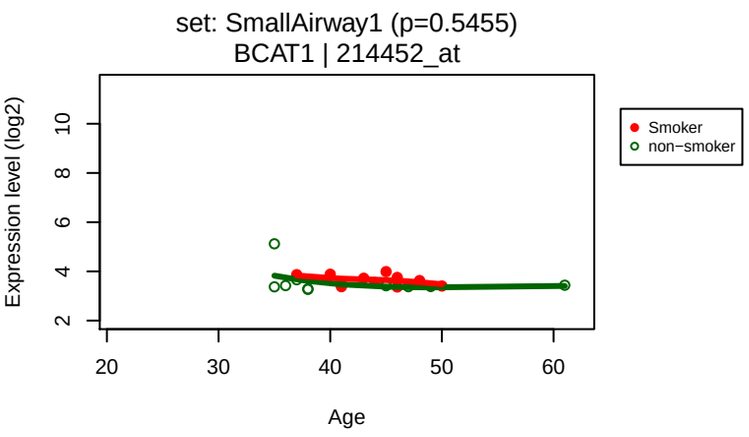
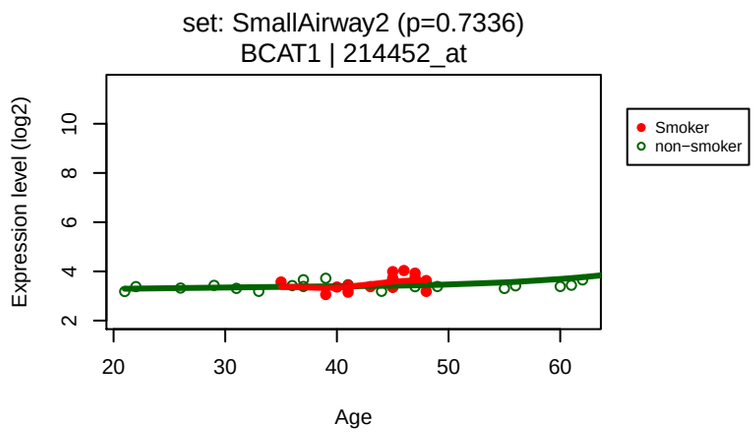
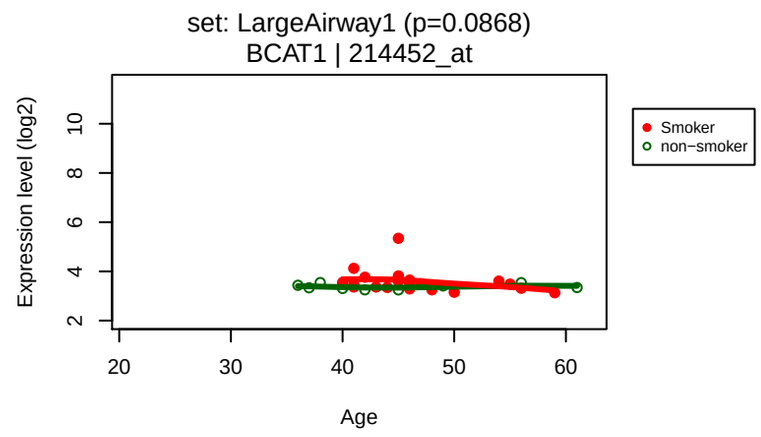

Gene expression of smokers and nonsmokers
gene: BCAT1 | 214452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5455
p=0.7336
p=0.557
p=0.0868
p=0.4505
p=0.0897
p=0.1814
p=0.4241
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

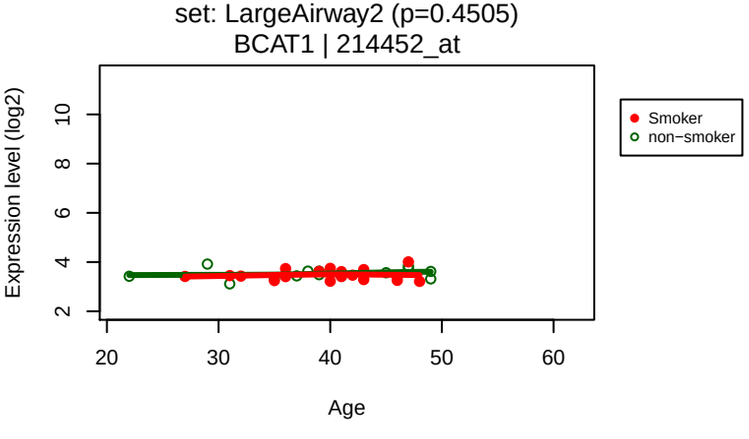

Gene expression ratio (smokers/nonsmokers)
gene: BCAT1 | 214452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5455
p=0.7336
p=0.557
p=0.0868
p=0.4505
p=0.0897
p=0.1814
p=0.4241
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

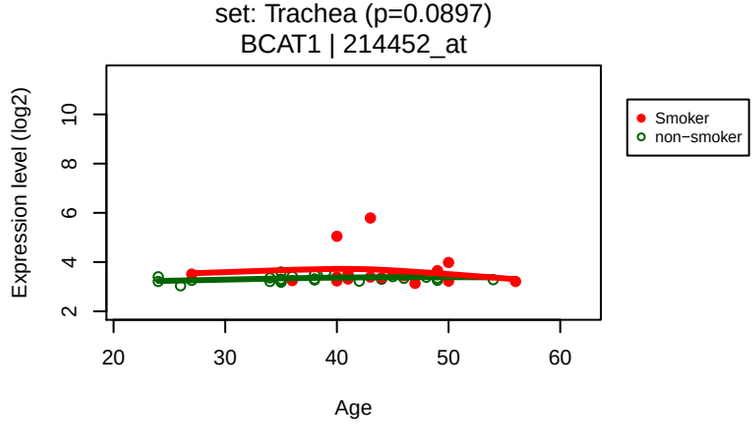

Gene expression ratio in Males
gene: BCAT1 | 214452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9915
p=0.6874
p=0.1193
p=0.4158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAT1 | 214452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5102
p=0.0394
p=0.5108
p=0.8864
SmallAirway2
LargeAirway2
Trachea
Macrophage2

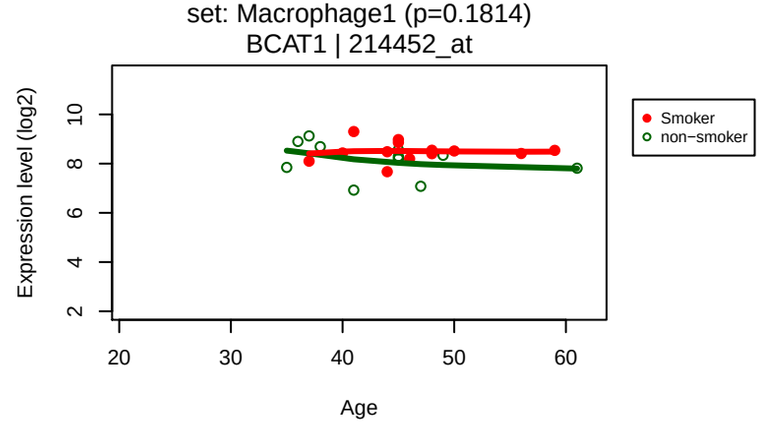

set: SmallAirway1 (p=0.5455)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker

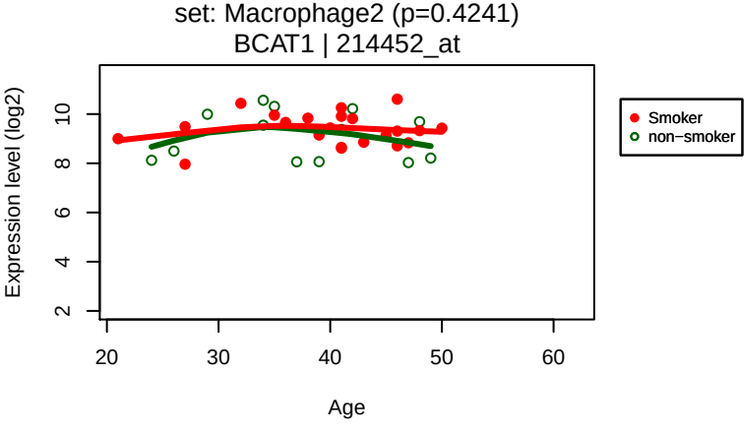

set: SmallAirway2 (p=0.7336)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker

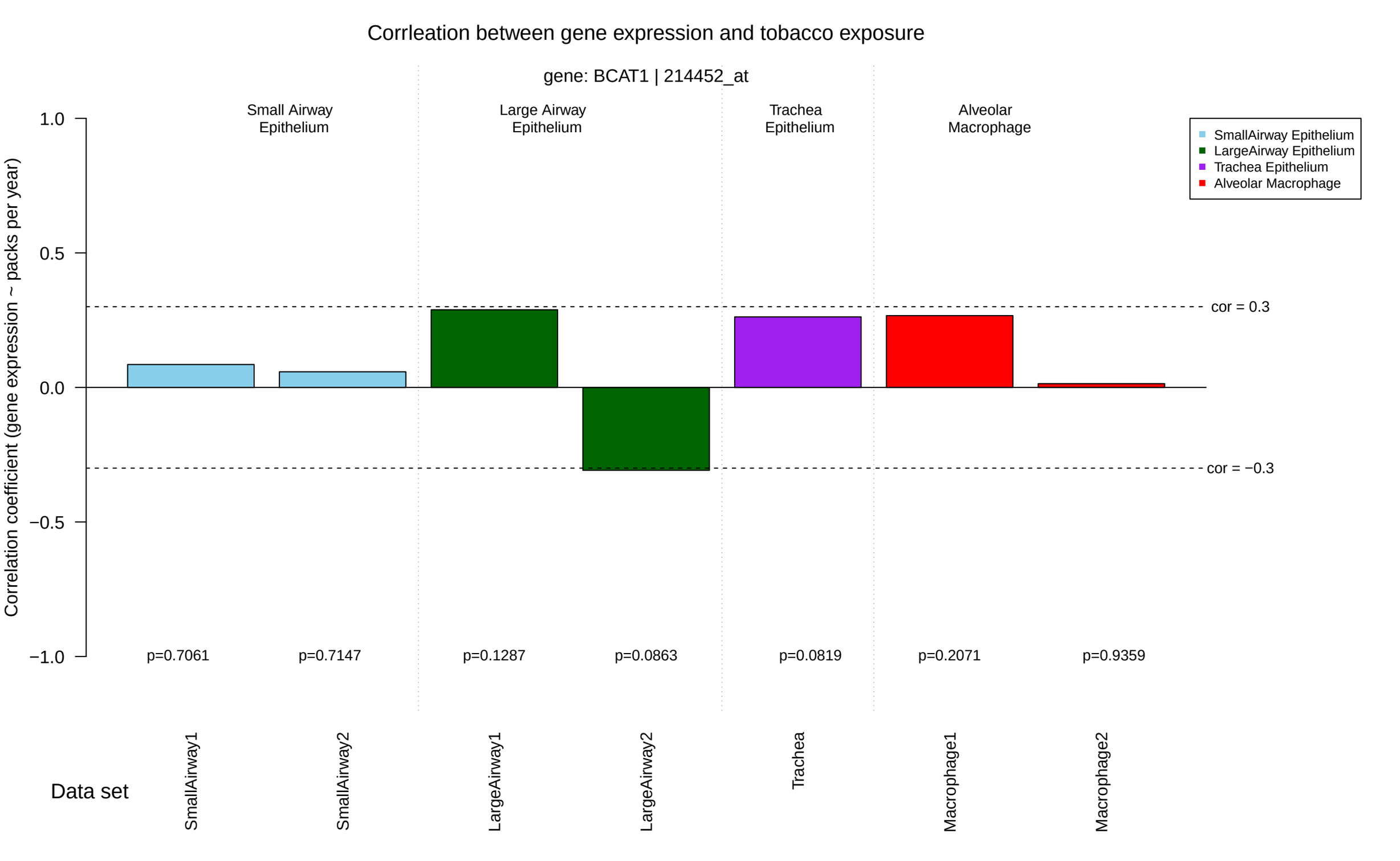

set: LargeAirway1 (p=0.0868)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4505)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0897)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1814)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4241)
BCAT1 | 214452_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAT1 | 214452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7061
p=0.7147
p=0.1287
p=0.0863
p=0.0819
p=0.2071
p=0.9359
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2